

Final Design Report
EK301 Section A1
Professor James G. McDaniel

Group 20: Truss Issues
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Introduction

The motivation behind our final truss design was to create a structure that maximizes load-carrying capacity while complying with the project's material, geometric, and cost constraints. After completing the preliminary design phase and reviewing feedback, our team focused on developing a layout that effectively distributes compressive forces, minimizes the likelihood of early buckling, and uses the limited acrylic material as efficiently as possible. Because buckling is the dominant failure mode for this project, we came up with a symmetric design with two elevated load joints and diagonals that shorten critical compression members and place the longer ones primarily in tension. Since the geometry was crucial, by examining multiple triangular arrangements and using our computational model to compare predicted failure loads, we identified structure designs that reduced compressive demand, maintained stiffness, and carried maximum load. Through this process, we agreed on a final design that balances strength, efficiency, and practical constructability.

Procedure

Since the Preliminary Design Report, we refined our design procedure in several important ways as we transitioned from initial concepts to the final optimized truss structure. In the preliminary phase, we evaluated two candidate truss geometries, each with distinct force distributions and overall configurations. Both preliminary designs revealed structural concerns during computational analysis, including several long compression members and some members (m_3 for Design 1, and m_1 and m_3 for Design 2) that did not meet the project length specifications ($8.5 \text{ in} \leq L_{jj} \leq 14 \text{ in}$). These outcomes indicated that our design procedure needed improvement, particularly in how we selected and adjusted geometric layouts.

After comparing the results of both preliminary designs, we chose to continue with Design 2 because it demonstrated a more balanced force distribution and more efficient load transfer than Design 1. Instead of creating a completely new geometry, our refined procedure focused on adjusting member lengths and shifting joint coordinates within the existing design framework to ensure compliance with all required project constraints: a 32-in overall truss span between the pin and roller supports, 2-in elevation of the load joint, individual member lengths within 8.5–14 in, and a maximum total member length of 120 in (see Figure 1).

To improve performance, we used an iterative optimization process where we made small geometric adjustments, ran our MATLAB analysis after each modification, and examined the effects on internal forces, buckling behavior, and total length. This approach allowed us to steadily reduce compressive force magnitudes, increase the buckling capacity of the most vulnerable members, and raise the predicted failure load while remaining within all project specifications. Rather than comparing completely different truss concepts, the procedure evolved into systematically refining a single design until it achieved a balanced geometry, acceptable member lengths, and improved load-bearing efficiency.

Analysis

The analytical approach for our final design used the same MATLAB truss-analysis program developed for the preliminary design (Appendix A.2). As in the earlier report, the program constructs the truss geometry from the specified joint coordinates and connection matrix, assembles the global equilibrium matrix including member direction cosines and support reactions, and solves for all internal forces under the required 32 oz applied load. This provides both the magnitude of each member force and whether the member is in tension or compression.

For compression members, we applied the same classwide buckling-fit relationship used in the preliminary design, $P_{crit}(L) = 1909.7L^{-1.74} \pm 1.58 \text{ oz}$, obtained from the Buckling Lab data.

The program computes the predicted buckling load of each compression member using this fit and applies the $\pm 1.58 \text{ oz}$ fit uncertainty (from class data) to account for experimental variation in the model. The compression member with the smallest predicted allowable load is identified as the critical member, and the maximum supported load of the truss is taken directly from this critical buckling value, with the same $\pm 1.58 \text{ oz}$ uncertainty carried into the final prediction. The theoretical failure load (maximum theoretical load) was then determined by scaling the applied 32 oz load until the internal force in the critical member reached each of these three buckling strengths. Since the internal forces scale linearly with load, we could calculate the nominal failure load directly from the ratio of buckling strength to member force.

The total cost of the truss was evaluated using the required cost equation $C = 1L_{total} + 10J$, based on total member length and number of joints. In addition to computing forces, buckling capacity, cost, and uncertainty, the program generates a plot of the truss that displays tension, compression, and zero-force members and highlights the critical member for clarity. A complete member table listing lengths, internal forces, and buckling predictions is also automatically produced.

Results

Final Design Diagram

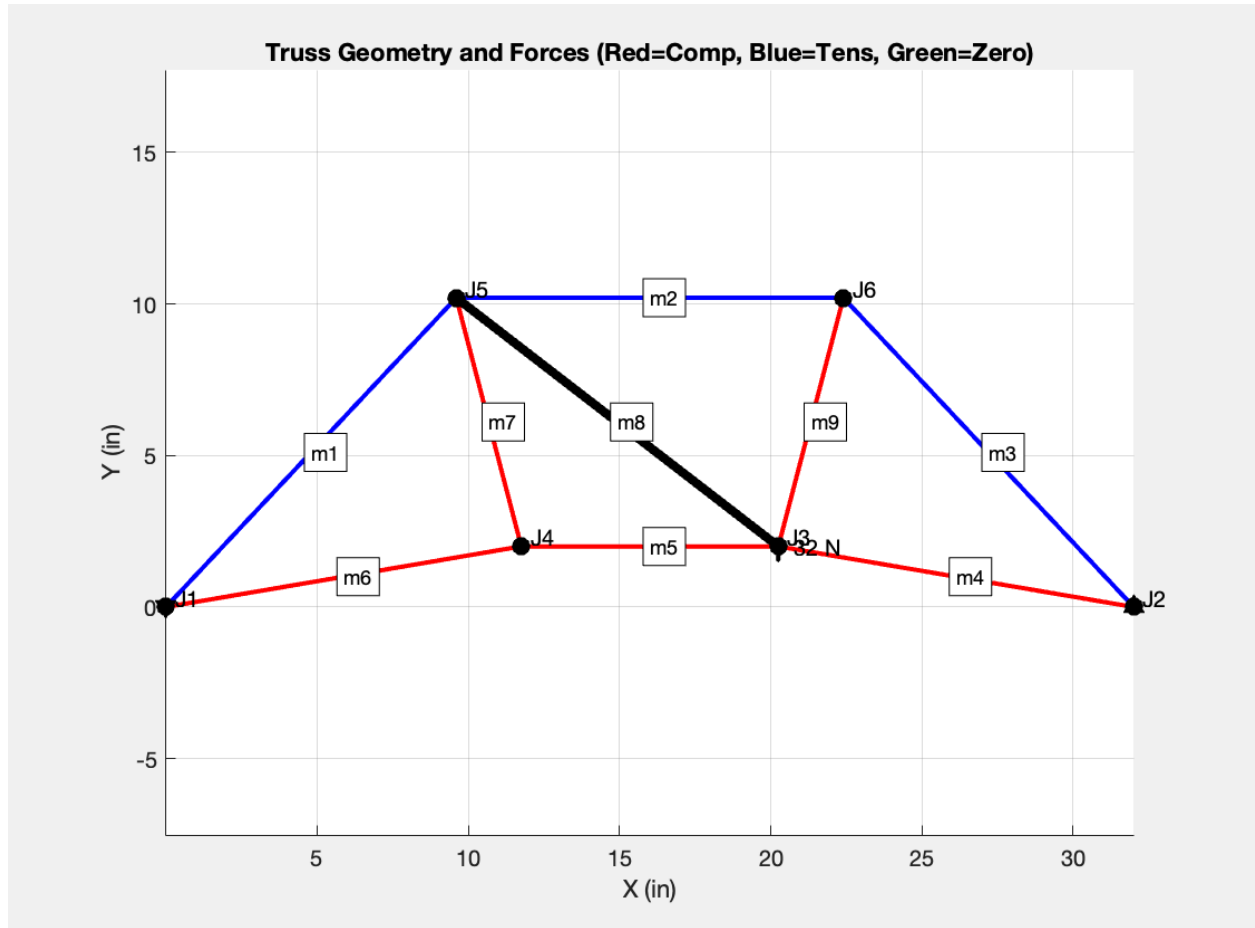


Figure 1. Final Design Diagram

Table 1: Final Design Members

Member	Length (in)	Type	Buckling Strength (oz)	Buckling Uncertainty (oz)	Force At Max Load (oz)
1	14.0	Tension	NaN	NaN	20.72
2	12.8	Tension	NaN	NaN	31.3
3	14.0	Tension	NaN	NaN	35.71
4	11.9	Compression	25.6	1.58	24.83
5	8.5	Compression	46.11	1.58	14.84
6	11.9	Compression	25.6	1.58	14.41
7	8.5	Compression	46.32	1.58	2.5
8	13.4	Compression	20.77	1.58	20.77
9	8.5	Compression	46.32	1.58	26.89

```
=== EK301 Truss Analysis (Cambered Design) ===

Saved input file: TrussDesign2_AscanioCerenAbdulahleef.mat

EK301, Section A1, Group 20: Ascanio Andreis, Ceren Atalar, Abdoulhaleef Sotunbo
Joints: 6 Members: 9
Total Length: 103.55 in
Cost: $163.55
Applied Load: 32.00 oz

Member Forces (oz):
  m1 :   19.214 (T)
  m2 :   29.017 (T)
  m3 :   33.113 (T)
  m4 :   23.021 (C)
  m5 :   13.756 (C)
  m6 :   13.358 (C)
  m7 :    2.317 (C)
  m8 :   19.260 (C)
  m9 :   24.928 (C)

Reactions (oz):
  Sx1 = 0.000
  Sy1 = 11.750
  Sy2 = 20.250

Max Theoretical Load = 34.51 oz
Load/Cost Ratio = 0.21103 oz/$

Critical Member: m8 (Compression)

Plot generated.
```

Figure 2. Final Design Command Window Output

Summary

Critical member (bold black): m8

- Length: 13.4 in
- Buckling Strength: 20.77 ± 1.58 oz

Max Theoretical Load: 34.51 ± 1.58 oz

Truss Cost: \$163.55

Load/Cost Ratio: 0.21103 oz/\$

Discussion & Conclusion

Our final truss design was driven by the goal of maximizing load capacity while keeping the structure efficient and within project constraints. We optimized the design by adjusting member lengths and joint positions, using computational analysis to see how each change affected force distribution. A key strategy was ensuring that the longest members carried tension rather than compression, since buckling in compression members limits performance.

The design evolved from comparing two preliminary concepts, and we built on Design 2 because it provided a better load bearing capacity and simpler geometry. Throughout the project, the most valuable components were learning how geometry affects internal forces and using the buckling model to guide improvements. If we were to revise the process, we would explore more variations in joint locations early on and perform sensitivity checks to understand how small geometric adjustments influence failure load. Overall, the project emphasized how crucial detailed modeling and geometry selection are in developing a strong and efficient truss.

Appendix

A.1 Hartford roof collapse discussion

Date and Time: 12/01/2025 7.30PM

Location: Zoom

Participants: Ascanio, Ceren, Haleef

Chair: Ascanio

Recorder: Ceren

Agenda

1. Summary of the Hartford Civic Center Arena Failure
 - Discuss key technical causes
 - Discuss procedural/ethical failures
 - Identify lessons relevant to our own design process
2. Discussion: Use of Computer Programs in Structural Analysis
 - Reliability of computer outputs (importance of understanding the model rather than trusting software blindly)
 - Sources of error
 - How we validated our own truss model (hand calculations, comparison to expected beam/truss behavior)
3. Determining an Appropriate Factor of Safety (FoS) for Our Truss
 - Typical FoS ranges for structures involving human safety (e.g., 2.0–3.0+)
 - Variability considerations:
 - Material imperfections, construction variability, load uncertainty, etc.
 - Discuss what FoS we will apply and justify it using ethical principles
4. Maximum Load Claim for Our Truss
 - Review experimental load data and theoretical predictions and decide on a conservative load capacity based on chosen FoS
5. Ethical Considerations
 - Relating decisions to NSPE Canon 1
 - How ethical failures contributed to the Hartford collapse and how we avoid similar mistakes

Meeting Minutes

Attendance: All members Present

Ascanio reviewed the planned agenda. The team agreed to proceed without changes.

1. Review of the Hartford Civic Center Collapse

Haleef summarized the major technical failures: long unbraced compression members, underestimated dead load, and missing midpoint bracing. He emphasized how many members were overstressed by hundreds of percent.

Ceren added that significant deflections were observed during construction but ignored.

Ascanio highlighted that buckling modes not included in the computer analysis were a major issue and that the engineers assumed the software couldn't be wrong.

The team agreed that the collapse showed a combination of design errors, construction deviations, and ethical complacency.

2. Use of Computer Programs in Structural Analysis

Ceren noted that in our own project we used online truss analysis software and that it was easy to misinterpret boundary conditions.

Ascanio added that software only calculates based on what you input; it doesn't warn you if you model the structure incorrectly. He said, "The Hartford engineers trusted the computer too much, we can't do that."

Haleef proposed that every software output in our report should be paired with either hand calculations or reasoning that supports why the result makes sense.

3. Appropriate Factor of Safety for Our Truss

Ascanio suggested an FoS of 2.0–2.5 since our truss is small but the hypothetical scenario involves human safety.

Ceren argued for 3.0, noting that material variability is significant and unpredictable.

Haleef said that since the assignment asks what we would claim if human life were at risk, we should stay on the conservative side.

The team agreed on a final Factor of Safety = 3.0.

4. Maximum Load Claim for Our Truss

Ceren estimated failure around 34.51 based on the analysis model.

Haleef recommended dividing that value by 3.0 to get the allowable load.

Ascanio stated that the claim has to be something we can defend ethically. Hartford failed partly because they overstated what the structure could safely handle.

The team agreed to claim a maximum safe load of:

$$\text{Allowable Load} = 34.51 / 3.0 = 11.503$$

5. Ethical Considerations

Ceren commented that the Hartford engineers ignored multiple warnings, which violated Canon 1.

Haleef said that even as students, we should reflect that safety matters more than making the numbers look good.

Ascanio emphasized noting in the report that ethical engineering requires questioning your own assumptions.

6. Conclusions

- $FoS = 3.0$
- maximum safe load = allowable load = failure load / 3.0 = $34.51 / 3.0 = 11.503$
- The team will emphasize verification of computer analysis.
- All ethical considerations will be tied back to NSPE Canon No. 1.

A.2 Truss Analysis Code for Final Design (Including Input File)

```
clear; clc; close all;
fprintf('=== EK301 Truss Analysis (Cambered Design) ===\n\n');

%% ==== USER INPUT SECTION ====
% Truss Geometry: 12 Joints (J) and 21 Members (M).
% Index Map: [A(1), C(2), E(3 - Load), D1(4), D2(5), B1(6), B2(7), B3(8),
B4(9), B5(10), B6(11), BC(12)]
X = [0, 32, 20.25, 11.75, 9.6, 22.4]; % in inches
Y = [0, 0, 2, 2, 10.2, 10.2]; % in inches
% Connection matrix (6 joints x 9 members) - FIXED: Member 9 connection added
to J9.
C = [
1,0,0,0,0,1,0,0,0;
0,0,1,1,0,0,0,0,0;
0,0,0,1,1,0,0,1,1;
0,0,0,0,1,1,1,0,0;
1,1,0,0,0,0,1,1,0;
0,1,1,0,0,0,0,0,1;
];
% Supports - 3 total reactions (pin = 2, roller = 1)
J = length(X);
R = 3; % Number of reactions
Sx = zeros(J,R);
Sy = zeros(J,R);
Sx(1,1) = 1; % Pin (x reaction) at Joint 1
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Sy(1,2) = 1;    % Pin (y reaction) at Joint 1
Sy(2,3) = 1;    % Roller (y reaction) at Joint 2 (C)
% External load vector
L = zeros(2*J,1);
W_load = 32;          % Oz.
load_joint = 3;       % applied at Joint 3 (E)
L(J + load_joint) = -W_load; % downward load in y
% Save to input file
input_file = 'TrussDesign2_AscanioCerenAbdulahleef.mat';
save(input_file, 'C', 'Sx', 'Sy', 'X', 'Y', 'L');
fprintf('Saved input file: %s\n\n', input_file);

%% ===== ANALYSIS SECTION =====
load(input_file);
J = length(X);
M = size(C,2);
R = size(Sx,2);    % Should be 3 reactions
% Determinacy check
if 2*J ~= (M + R)
    fprintf('Warning: Truss is indeterminate (2J = %d, M+R = %d).\n', 2*J, M+R);
end
% Build equilibrium matrix A (2J × (M + R))
A = zeros(2*J, M + R);
% Part 1: Member direction cosines
for j = 1:J
    for m = 1:M
        if C(j,m) == 1
            k = find(C(:,m) == 1);
            k = k(k ~= j);
            dx = X(k) - X(j);
            dy = Y(k) - Y(j);
            Lm = sqrt(dx^2 + dy^2);
            A(j, m) = dx / Lm;
            A(J+j, m) = dy / Lm;
        end
    end
end
% Part 2: Reaction forces
A(:, M+1:end) = [Sx; Sy];
% Solve system
T = A \ L;
member_forces = T(1:M);
reactions = -T(M+1:end);

%% = LENGTHS / COST / BUCKLING =
member_lengths = zeros(M,1);
for m = 1:M
    idx = find(C(:,m));
    dx = X(idx(2)) - X(idx(1));

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dy = Y(idx(2)) - Y(idx(1));
member_lengths(m) = sqrt(dx^2 + dy^2);
end
total_length = sum(member_lengths);
C1 = 10; % per joint
C2 = 1; % per inch
cost = C1*J + C2*total_length;
% Buckling
C_buck = 1909.7; alpha = 1.74;
Pcrit = C_buck * member_lengths.^(-alpha);
comp_idx = find(member_forces < 0);
if isempty(comp_idx)
W_failure = Inf;
else
Rc = member_forces(comp_idx) / W_load;
Pcrit_c = Pcrit(comp_idx);
W_failure = min(-Pcrit_c ./ Rc);
end
cost_ratio = W_failure / cost;

%% ===== RESULTS =====
fprintf('EK301, Section A1, Group 20: Ascanio Andreis, Ceren Atalar,
Abdoulhaleef Sotunbo\n');
fprintf('Joints: %d Members: %d\n', J, M);
fprintf('Total Length: %.2f in\n', total_length);
fprintf('Cost: $%.2f\n', cost);
fprintf('Applied Load: %.2f oz\n\n', W_load);
fprintf('Member Forces (oz):\n');
for m = 1:M
type = 'T';
if member_forces(m) < 0, type = 'C';
elseif abs(member_forces(m)) < 1e-4, type = 'Z';
end
fprintf(' m%-2d: %8.3f (%s)\n', m, abs(member_forces(m)), type);
end
fprintf('\nReactions (oz):\n');
rx_labels = ["Sx1", "Sy1", "Sy2"];
for r = 1:R
fprintf(' %s = %.3f\n', rx_labels(r), reactions(r));
end
fprintf('\nMax Theoretical Load = %.2f oz\n', W_failure);
fprintf('Load/Cost Ratio = %.5f oz/$\n', cost_ratio);

%% ===== PLOTTING SECTION =====
figure('Name', 'Truss Plot'); hold on; axis equal;
title('Truss Geometry and Forces (Red=Comp, Blue=Tens, Green=Zero)');
xlabel('X (in)'); ylabel('Y (in)');
grid on;
max_abs_force = max(abs(member_forces));

```

```

force_scale = 2 / max_abs_force;
% Determine critical member
if ~isempty(comp_idx)
    [~, local_idx] = min(-Pcrit(comp_idx) ./ (member_forces(comp_idx)/W_load));
    crit_member = comp_idx(local_idx);
    fprintf('\nCritical Member: m%d (Compression)\n', crit_member);
else
    fprintf('\nNo compression members - no critical member.\n');
end
% Plot members
for m = 1:M
    idx = find(C(:,m));
    x = X(idx);
    y = Y(idx);
    force = member_forces(m);
    if abs(force) < 1e-4
        col = 'g'; ls = '--'; lw = 1;
    elseif force > 0
        col = 'b'; ls = '-'; lw = 2;
    else
        col = 'r'; ls = '-'; lw = 2;
    end
    plot(x, y, 'Color', col, 'LineWidth', lw, 'LineStyle', ls);
    if exist('crit_member','var') && m == crit_member
        plot(x, y, 'k', 'LineWidth', 4); % highlight critical member
    end
    text(mean(x), mean(y),
    sprintf('m%d',m), 'FontSize', 9, 'BackgroundColor', 'w', 'EdgeColor', 'k');
end
% Joints
scatter(X, Y, 70, 'k', 'filled');
for j = 1:J
    text(X(j)+0.3, Y(j)+0.3, sprintf('J%d', j));
end
% Supports
plot(X(1), Y(1), 'kv', 'MarkerSize', 8, 'MarkerFaceColor', 'k'); % pin at Joint
1
plot(X(2), Y(2), 'k^', 'MarkerSize', 8, 'MarkerFaceColor', 'k'); % roller at
Joint 2
% Load arrow
quiver(X(load_joint), Y(load_joint), 0, -0.5, 'k', 'LineWidth', 2);
text(X(load_joint)+0.5, Y(load_joint), sprintf('%.0f N', W_load),
'FontSize', 10);
hold off;
fprintf('\nPlot generated.\n');

%% ==== MEMBER TABLE (CSV) ====
% Constants for buckling model
C_fit = 1909.7;

```

```

alpha_fit = 1.74;
U_fit = 1.58;    % Uncertainty (±1.58 oz)
% Preallocate
MemberNumber = (1:M).';
MemberLength = member_lengths;
ForceType = strings(M,1);
BucklingStrength = nan(M,1);
BucklingUncertainty = nan(M,1);
ForceAtMaxLoad = abs(member_forces) * (W_failure / W_load);
for m = 1:M
    if member_forces(m) < 0
        ForceType(m) = "Compression";
        % Buckling force from professor's model
        BucklingStrength(m) = C_fit * (member_lengths(m)^(-alpha_fit));
        % Constant fit uncertainty
        BucklingUncertainty(m) = U_fit;
    elseif member_forces(m) > 0
        ForceType(m) = "Tension";
    else
        ForceType(m) = "Zero-force";
    end
end
% Build MATLAB table
MemberTable = table( ...
    MemberNumber, ...
    round(MemberLength,1), ...
    ForceType, ...
    round(BucklingStrength,2), ...
    round(BucklingUncertainty,2), ...
    round(ForceAtMaxLoad,2), ...
    'VariableNames', { ...
        'Member', ...
        'Length_in', ...
        'Type', ...
        'BucklingStrength_oz', ...
        'BucklingUncertainty_oz', ...
        'ForceAtMaxLoad_oz' ...
    });
% Write CSV file
csv_filename = 'Truss_Member_Table.csv';
writetable(MemberTable, csv_filename);
fprintf('\nCSV table generated: %s\n', csv_filename);

```